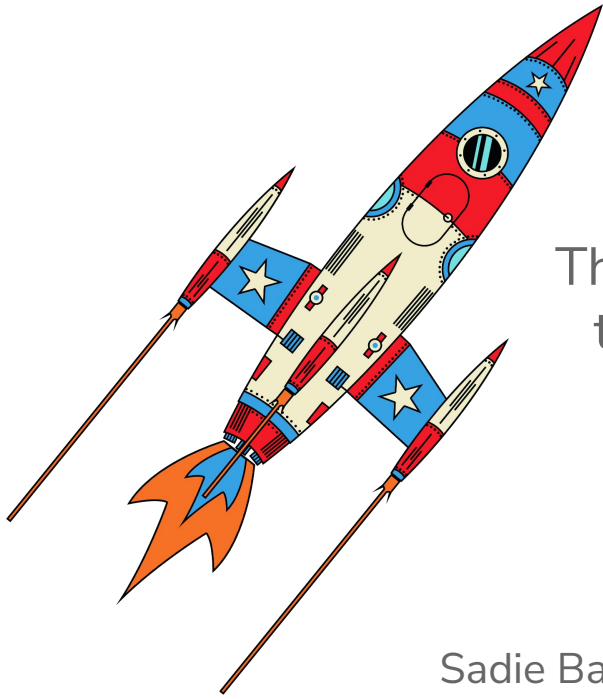




Up-Goer Five

The world of NCAS using only the
thousand (“ten hundred”) most
common words

1000

Sadie Bartholomew, NCAS-CMS & University of Reading
NCAS Staff Meeting 2026 | 2026-06-30 | Staverton, Daventry



**National Centre for
Atmospheric Science**

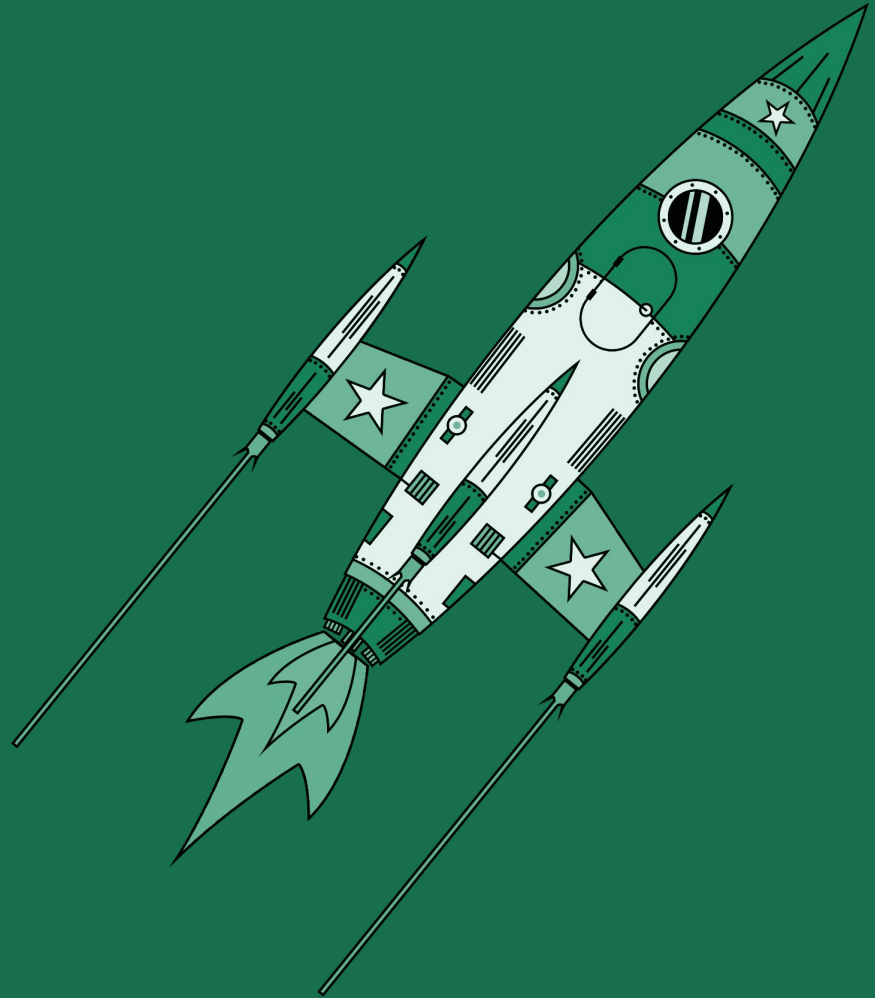
NATURAL ENVIRONMENT RESEARCH COUNCIL



**University of
Reading**

Part 1 of 3

Context



**National Centre for
Atmospheric Science**

NATURAL ENVIRONMENT RESEARCH COUNCIL

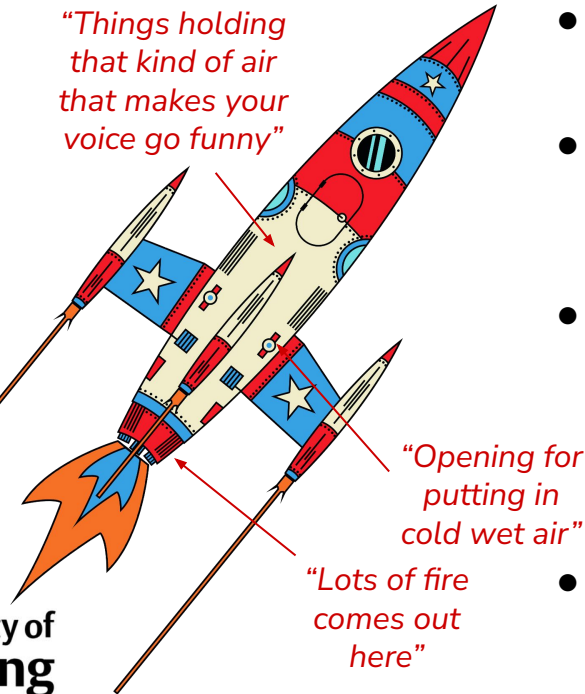
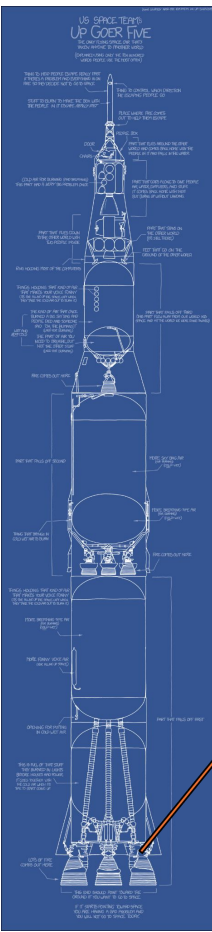
What is Up Goer Five (and the associated Challenge)?

Some definitions from the original comic: NASA Saturn V rocket i.e. "The only flying space car that's taken anyone to another world."

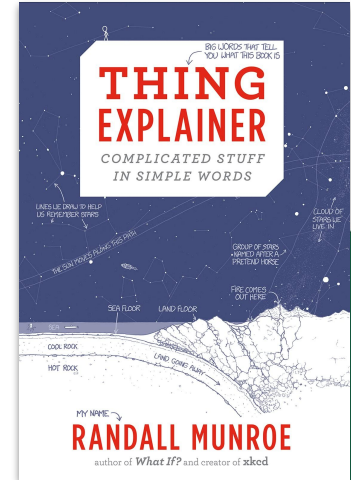
"Things holding that kind of air that makes your voice go funny"

"Opening for putting in cold wet air"

"Lots of fire comes out here"



- The name and idea comes from an XKCD web comic (#1133, <https://xkcd.com/1133/>, image to left but too tall to show as-is!) which describes parts of a rocket in terms of only the 100 most commonly used English words.
- The author Randall Munroe also wrote a book 'Thing Explainer' based on the concept
- Basic example: 'thousand' isn't in the top 1000 words, so we have to use 'ten hundred' instead
- The 'Up Goer Five Challenge' invites us to explain complex concepts or scientific research using only the 1,000 most common words in English
- A nice resource of examples: <https://tenhundredwordsofscience.tumblr.com/>

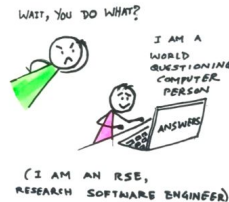


Why am I hosting a session about this?

- I first heard about Up Goer Five when attending the Software Sustainability Institute's (SSI) Collaborations Workshop 2026 (CW26), where in the Hack Day I joined a team whose project focused on defining terms related to the RSE career and community. We created a GitHub repository github.com/software saved/up-goer-five with an associated webpage softwaresaved.github.io/up-goer-five/ showing terms we defined (small portion shown below)
- I thought it would be nice to do a similar exercise for NCAS: our research/science, structure, job roles, etc.
- **Using very limited language is a useful exercise for all** because it challenges us to get to the very core of a concept, building communication skills. And the results are helpful too: in theory we can then explain the concept to *anyone*, because we have used only the *most basic building blocks*

Computer People Things

Explaining Work with Computers to Find Things Out



Research Software Engineer World Questioning Computer Person

I am a person who writes computer work to help find answers for a question. My work is to build and fix computer work used in studies. I make sure they work and share it with other people to use.



Research Community Manager World questioning people grouper

I lead a group of different people who work to understand the world or people by looking for new facts. My work is to get money to let us do this, and make sure that we do it in a good way so that others can learn from it.



Collaborations Workshop Lots of different people working together meeting

An often exciting and sometimes fun meeting bringing many different people to have interesting conversations to change the world and themselves



What are the 1000 words?

Note: • plurals, verb forms, comparatives, irregular forms, for each are all allowed • no words start with X or Z

A: a•able•about•above•accept•across•act•actually•add•admit•afraid•after•afternoon•again•against•age•ago•agree•ah•ahead•air•all•allow•almost•alone•along•already•alright•also•although•always•am•amaze•an•and•anger•angry•animal•annoy•another•answer•any•anybody•anyhow•anyone•anything•anyway•apartment•apparently•appear•approach•are•area•aren't•arm•around•arrive•as•ask•asleep•ass•at•attack•attempt•attention•aunt•avoid•away

B: baby•back•bad•bag•ball•band•bar•barely•bathroom•be•beat•beautiful•became•because•become•bed•bedroom•been•before•began•begin•behind•believe•bell•beside•besides•best•better•between•big•bit•bite•black•blink•block•blood•blue•blush•body•book•bore•both•bother•bottle•bottom•box•boy•boyfriend•brain•break•breakfast•breath•breathe•bright•bring•broke•broken•brother•brought•brown•brush•build•burn•burst•bus•business•busy•but•buy•by

C: call•calm•came•can•can't•car•card•care•carefully•carry•case•cat•catch•caught•cause•cell•chair•chance•change•chase•check•cheek•chest•child•children•chuckle•city•class•clean•clear•climb•close•clothes•coffee•cold•college•col•come•comment•complete•completely•computer•concern•confuse•consider•continue•control•conversation•cool•corner•couch•could•couldn't•counter•couple•course•cover•crack•crazy•cross•crowd•cry•cup•cut•cute

D: dad•damn•dance•dark•date•daughter•day•dead•deal•dear•death•decide•deep•definitely•desk•didn't•die•different•dinner•direction•disappear•do•doctor•does•doesn't•dog•don't•done•door•doubt•down•drag•draw•dream•dress•drink•drive•drop•drove•dry•during

E: each•ear•early•easily•easy•eat•edge•either•else•empty•end•enjoy•enough•enter•entire•escape•especially•even•evening•eventually•ever•every•everyone•everything•exactly•except•excite•exclaim•excuse•expect•explain•express•on•eye•eyebrow

F: face•fact•fall•family•far•fast•father•fault•favorite•fear•feel•feet•fell•felt•few•field•fight•figure•fill•finally•find•fine•finger•finish•fire•first•fit•five•fix•flash•flip•floor•fly•focus•follow•food•foot•for•force•forget•form•forward•found•four•free•fri•end•from•front•frown•fuck•full•fun•funny•further

G: game•gasp•gave•gaze•gently•get•giggle•girl•girlfriend•give•given•glad•glance•glare•glass•go•God•gone•gonna•good•got•gotten•grab•great•green•greet•grey•grin•grip•groan•ground•group•grow•guard•guess•gun•guy

H: had•hadn't•hair•half•hall•hallway•hand•handle•hang•happen•happy•hard•has•hate•have•haven't•he•he'd•he's•head•hear•heard•heart•heavy•held•hell•hello•help•her•here•herself•hey•hi•hide•high•him•himself•his•hit•hold•home•horse•hospital•hot•hour•house•how•however•hug•huge•huh•human•hundred•hung•hurry•hurt

I: I'd•I'll•I'm•I've•ice•idea•if•ignore•imagine•immediately•important•in•inside•instead•interest•interrupt•into•is•isn't•it•it's•its

J: jacket•jeans•jerk•job•join•joke•jump•just

K: keep•kept•key•kick•kid•kill•kind•kiss•kitchen•knee•knew•knock•know•known

L: lady•land•large•last•late•laugh•lay•lead•lean•learn•least•leave•led•left•leg•less•let•letter•lie•life•lift•light•like•line•lip•listen•little•live•lock•locker•long•look•lose•lost•lot•loud•love•low•lunch

M: mad•made•make•man•manage•many•mark•marry•match•matter•may•maybe•me•mean•meant•meet•memory•men•mention•met•middle•might•mind•mine•minute•mirror•miss•mom•moment•money•month•mood•more•morning•most•mother•mouth•move•movie•Mr. Mrs. much•mum•mumble•music•must•mutter•my•myself

N: name•near•nearly•neck•need•nervous•never•new•next•nice•night•no•nod•noise•none•normal•nose•not•note•nothing•notice•now•number

O: obviously•of•off•offer•office•often•oh•okay•old•on•once•one•only•onto•open•or•order•other•our•out•outside•over•own

P: pack•pain•paint•pair•pants•paper•parents•park•part•party•pass•past•pause•pay•people•perfect•perhaps•person•phone•pick•picture•piece•pink•piss•place•plan•play•please•pocket•point•police•pop•position•possible•power•practically•present•press•pretend•pretty•probably•problem•promise•pull•punch•push•put

Q: question•quick•quickly•quiet•quietly•quite

R: race•rain•raise•ran•rang•rather•reach•read•ready•real•realize•really•reason•recognize•red•relationship•relax•remain•remember•remind•repeat•reply•respond•rest•return•ride•right•ring•road•rock•roll•room•rose•round•rub•run•rush

S: sad•safe•said•same•sat•save•saw•say•scare•school•scream•search•seat•second•see•seem•seen•self•send•sense•sent•serious•seriously•set•settle•seven•several•shadow•shake•share•she•she'd•she's•shift•shirt•shit•shock•shoe•shook•shop•short•shot•should•shouldn't•shout•shove•show•shower•shrug•shut•sick•side•sign•sight•sign•silence•silent•simply•since•single•sir•sister•sit•situation•six•skin•sky•slam•sleep•slightly•slip•slow•slowly•small•smell•smile•smirk•smoke•snap•so•soft•softly•some•somehow•someone•something•sometimes•somewhere•son•song•soon•sorry•sort•sound•space•speak•spend•spent•spoke•spot•stair•stand•star•stare•start•state•stay•step•stick•stomach•stood•stop•store•story•straight•strange•street•strong•struggle•stuck•student•study•stuff•stupid•such•suck•sudden•suddenly•suggest•summer•sun•suppose•sure•surprise•surround•sweet

T: table•take•taken•talk•tall•teacher•team•tear•teeth•tell•ten•than•thank•that•that's•the•their•them•themselves•then•there•there's•these•they•they'd•they're•thick•thing•think•third•this•those•though•thought•three•threw•throat•through•throw•tie•tight•time•tiny•tire•to•today•together•told•tomorrow•tone•tongue•tonight•too•took•top•totally•touch•toward•town•track•trail•train•tree•trip•trouble•trust•true•truth•try•turn•TV•twenty•two•type

U: uncle•under•understand•until•up•upon•us•use•usual•usually

V: very•visit•voice

W: wait•wake•walk•wall•want•warm•warn•was•wasn't•watch•water•wave•way•we•we'll•we're•we've•wear•week•weird•well•went•were•weren't•wet•what•what's•whatever•when•where•whether•which•while•whisper•white•who•whole•why•wide•wife•will•wind•window•wipe•wish•with•within•without•woke•woman•women•won't•wonder•wood•word•work•world•worry•worse•would•wouldn't•wow•wrap•write•wrong

Y: yeah•year•yell•yes•yet•you•you'd•you'll•you're•you've•young•your•yourself



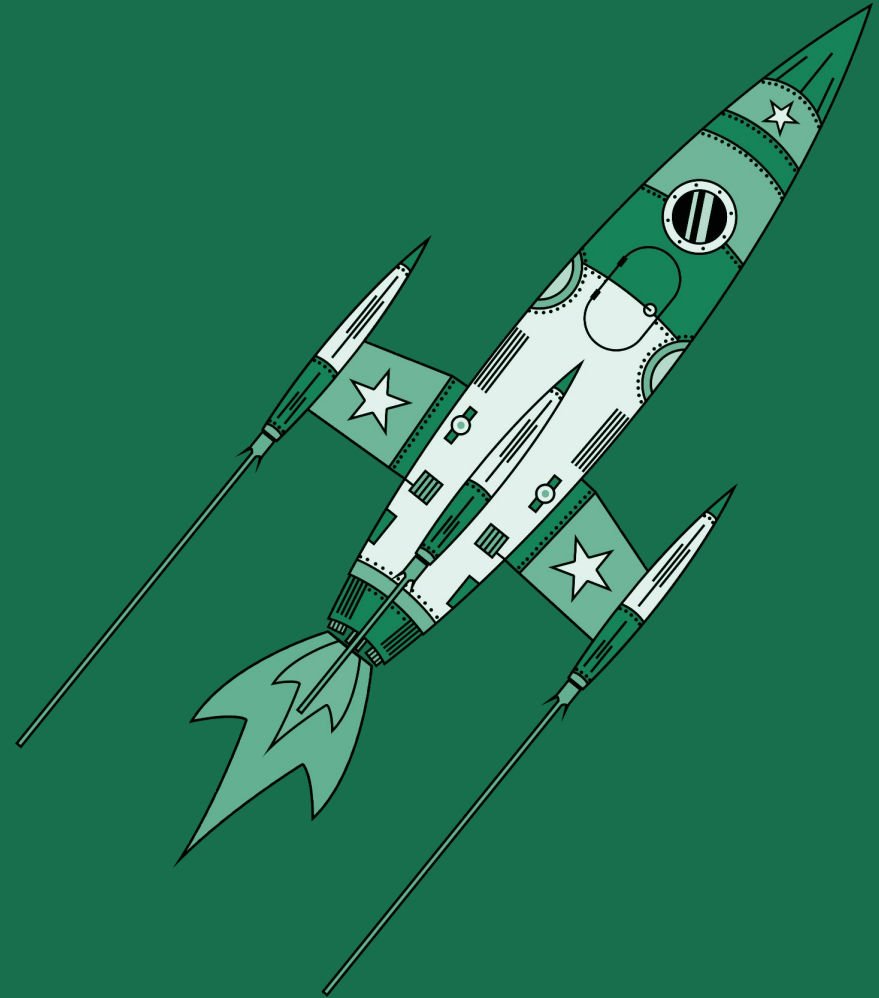
The Up Goer Five Challenge in (earth & aligned) science

- This challenge has been discussed in particular for a research and science context e.g. a good article is: <https://eos.org/articles/can-you-explain-science-using-only-1000-common-words>
- Some sessions have been held at big conferences in the domain e.g. AGU session in 2018 (<https://agu.confex.com/agu/fm18/meetingapp.cgi/Session/59282>). Examples from that include:
 1. Megan Johnson and Fernando Garcia Menendez summarising wildfires and their impacts:
“Smoke from crazy fires, or fires that are not planned, makes the air not clean. Smoke can make people sick—especially those that are young, old, or already sick—or even cause them to die early.”
 2. Frederick Cheng and Nandita Basu summarising how agricultural runoff and pollutants induce endocrine disruption and sex reversal in fish:
“People really like to throw away stuff and this stuff usually ends up in our water. So now a lot of our water is full of bad things. Sometimes, these bad things will cause big problems for fish and people. One of these bad things can actually turn boy fish into girl fish if they are in the not-clean water for a long time!”
- And some illustrative titles of talks from that session:
 1. Rubbing and pressing rocks together with a big computer controlled press to learn about the times when the ground shakes a lot as pieces of the world move around
 2. Why Are the Pieces of Land in the Wide Water that Breath out Fire and Smoke Made of Different Types of Rocks?
 3. How can we make more power by making air take up less space, putting it under the ground, and then bringing it back up?
 4. Are Sky Long Water Things Warming Up?



Part 2 of 3

Over to you!



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Tools and references to help with the Up Goer Five challenge

The NCAS website 'our role' description, in case you don't recognise the example text!

- Picking out words from the list is quite tedious and slow, so use tools to help us!
www.xkcd.com/simplewriter/ will validate any text, highlighting 'bad' words in red:

xkcd **SIMPLE WRITER**
WRITE LIKE UP GOER FIVE AND THING EXPLAINER

PUT WORDS HERE

We provide the UK with the knowledge and expertise to remain at the forefront of atmospheric science.

Our research, skills and infrastructure lead to climate change resilience, clean air solutions, and early-warning systems for high-risk weather.

- <https://web.archive.org/web/20260414215130/https://splasho.com/upgoer5/phpspellcheck/dictionaries/1000.dicin> lists all of the words, if useful
- Arrange yourself so you are on a table where at least one person has a laptop and accesses this validator tool - ideally as many people as possible can use it for instant feedback on whether a definition uses only the 1000 words



Reference: the 1000 words and some potentially more useful inclusions

- Left: full list (as on handout, but use validator if can); right: more useful(?) words from the 1000

A: a·able·about·above·accept·across·act·actually·add·admit·afraid·after·afternoon·again·against·age·ago·agree·ah·ahead·air·all·allow·almost·alone·along·a ready·alright·also·although·always·am·amaze·an·and·anger·angry·animal·annoy·another·answer·any·any more·any one·anything·any way·apartment·appear
nly·appear·approach·are·area·aren't·arm·arrive·as·ask·asleep·ass·at·attack·attempt·attention·aunt·avoid·away
B: b·baby·back·bad·bag·ball·band·bar·barely·bathroom·be·beat·beautiful·became·because·become·bed·bedroom·been·before·began·begin·behind·believe·b ell·beside·besides·best·better·between·big·bit·bite·black·blink·block·blonde·blood·blue·blush·body·book·bore·both·bother·bottom·box·boy·boyfriend·b rain·break·breakfast·breath·breathe·bright·bring·broke·broken·brother·brought·brown·brush·build·burn·burst·bus·business·busy·but·buy·by
C: call·calm·came·can·can't·car·card·care·carefully·carry·case·cat·catch·caught·cause·cell·chair·chance·change·check·cheek·chest·child·children·ch uckle·city·class·clean·clear·climb·close·clothes·coffee·cold·college·color·come·comment·complete·completely·computer·concern·confuse·consider·continue
·control·conversation·cool·corner·couch·could·couldn't·counter·couple·course·cover·crack·crazy·cross·crowd·cry·cup·cut·cute
D: dad·damn·dance·dark·date·daughter·day·dead·deal·death·decide·deep·definitely·desk·did·didn't·die·different·dinner·direction·disappear·do·doctor·d oes·doesn't·dog·don't·done·door·doubt·down·drag·draw·dream·dress·drink·drive·drop·drove·dry·during
E: each·ear·early·easily·easy·eat·edge·either·else·empty·end·enjoy·enough·enter·entire·escape·especially·even·evening·eventually·ever·every·everyone·ev erything·exactly·except·excite·exclaim·excuse·expect·explain·expression·eye·eyebrow
F: face·fact·fall·family·far·fast·father·fault·favorite·fear·feel·felt·fell·felt·few·field·fight·figure·fill·finally·find·fine·finger·finish·fire·first·fit·five·fix·flash·flip·floor·fl y
·focus·follow·food·foot·for·force·forget·form·forward·found·four·free·friend·from·front·frown·fuck·full·fun·funny·further
G: game·gasp·gave·gaze·gently·get·giggle·girl·girlfriend·give·given·glad·glance·glare·glass·go·God·gone·gonna·good·got·gotten·grab·great·green·greet·gre y·grin·grip·groan·ground·group·grow·guard·guess·gun·gunny
H: had·hadn't·hair·half·hall·halfway·hand·handle·hang·happen·happy·hard·has·hate·have·haven't·he·he'd·he's·head·hear·heard·heart·heavy·held·hell·hello· help·her·here·herself·hey·hi·hide·high·him·himself·his·hit·hold·home·hope·horse·hospital·hot·hour·house·how·however·hug·huge·huh·human·hundred·hun g·hurry·hurt
I: I'd·I'll·I'm·I've·ice·idea·if·ignore·imagine·immediately·important·in·inside·instead·interest·interrupt·into·is·isn't·it·it's·its
J: jacket·jeans·jerk·job·join·joke·jump·just
K: keep·kept·key·kick·kid·kill·kind·kiss·kitchen·knee·knew·knock·know·known
L: lady·land·large·last·late·laugh·lay·lead·lean·learn·least·leave·left·leg·less·let·letter·lie·life·lift·light·like·line·lip·listen·little·live·lock·locker·long·look·lose· lost·lot·loud·love·low·lunch
M: mad·made·make·man·manage·many·mark·marry·match·matter·may·maybe·me·mean·meant·meet·memory·men·mention·met·middle·might·mind·mine· minute·mirror·miss·mom·moment·money·month·mood·more·morning·most·mother·mouth·move·movie·Mr·Mrs·much·mum·mumble·music·must·mutter·my
·myself
N: name·near·nearly·neck·need·nervous·never·new·next·nice·night·no·nod·noise·none·normal·nose·not·note·nothing·notice·now·number
O: obviously·off·offer·office·often·oh·okay·old·on·once·one·only·onto·open·or·order·other·our·out·outside·over·own
P: pack·pain·paint·pair·pants·paper·parents·park·part·party·pass·past·pause·pay·people·perfect·perhaps·person·phone·pick·picture·piece·pink·piss·place·pl ac·play·please·pocket·point·police·pop·position·possible·power·practically·present·press·pretend·pretty·probably·problem·promise·pull·punch·push·put
Q: question·quick·quickly·quiet·quietly·quite
R: race·rain·raise·ran·rang·rather·reach·read·ready·real·really·reason·recognize·red·relationship·relax·remain·remember·remind·repeat·reply·respon d·rest·return·ride·right·ring·road·rock·roll·room·rose·round·rub·run·rush
S: sad·safes·safe·same·sat·save·saw·say·scares·school·scream·search·seat·second·see·seem·seen·self·send·sense·sent·serious·seriously·set·settle·seven· several·shadow·shakes·share·she·she's·shift·shirt·shit·shock·shoe·shop·short·shot·should·shouldn't·shout·shove·show·shower·shrug· shut·sick·side·sigh·sight·sign·silence·silent·simply·since·single·sir·sister·sit·situation·six·skin·sky·slam·sleep·slightly·slip·slow·slowly·small·smell·smile·smir k·smoke·snap·so·soft·softly·some·somehow·someone·something·sometimes·somewhere·son·song·soon·sorry·sort·sound·space·speak·spend·spent·spoke ·spot·stair·stand·star·stare·start·state·stay·step·stick·still·stomach·stood·stop·store·story·straight·stranger·street·strong·struggle·stuck·student·study·stuff·stu pid·suck·sudden·suddenly·suggest·summer·sun·suppose·sure·surprise·surround·sweet
T: table·take·taken·talk·tall·teacher·team·teeth·tell·ten·than·thank·that·that's·the·they·there·these·these they·they'd·they're·thic k·thing·think·third·this·those·though·thought·three·throw·throat·through·throw·tie·tight·time·tiny·tire·to·today·together·told·tomorrow·tone·tongue·tonight·too k·top·totally·touch·toward·town·track·trail·train·tree·trip·trouble·trust·true·truth·try·turn·TV·twenty·two·type
U: uncle·under·understand·until·upon·us·use·usual·usually
V: very·visit·voice
W: wait·wake·walk·wall·want·warm·warm·was·wasn't·watch·water·wave·way·we·we'll·we're·we've·we wear·week·weird·well·went·were·weren't·wet·what·what's
·whatever·when·where·whether·which·while·whisper·white·who·whole·why·wide·wife·will·wind·window·wipe·wish·with·within·without·woke·woman·women·w on't·won't·wood·word·wore·work·world·worry·worse·would·wouldn't·wow·wrap·write·wrong
Y: yeah·year·yell·yes·yet·you·you'd·you'll·you're·you've·young·your·yourself

Note: heading words not drawn from the list of 1000 words!

- **weather and climate (atmosphere etc.):** air, cold, dry, earth, fire, ice, rain, sky, snow, storm, sun, warm, water, wet, wind, light
- **science, research and understanding:** body, cause, change, fact, figure, find, idea, important, learn, matter, number, reason, study, thing, understand
- **data, computing and modelling:** answer, build, call, change, computer, control, machine, memory, number, point, power, problem, run, set, system, use, work, move
- **observation, measurement and analysis:** check, count, eye, fact, figure, find, hear, look, read, see, show, watch
- **prediction, time and change:** time, change, future, guess, happen, history, know, next, past, possible, question, time, understand
- **communication, education and public work:** ask, clear, help, message, say, speak, tell, word, write



A warm up: try it yourself

- To practice, let's all make our own definitions for the same words and see how they compare.
- I will put up a reference slide but refer to the sheets on your desk if helpful - and if you have laptop access use the validator tool which makes it simplest to check words.
- For 1 to 3, please be as creative as possible - it would be most interesting to get different definitions despite the limited vocabulary we can use!

Individually, create definitions using only the 1000 most common words (you have 5 minutes to cover them all) for:

1. Golf
2. Weather
3. Cloud (you may choose the context - meteorology or computing!)
4. Your own role and/or job title (e.g. 'scientist', 'technician', 'project/finance officer' etc.)



Main activity: describing the world of NCAS

- Now let's work together in tables to think about NCAS!

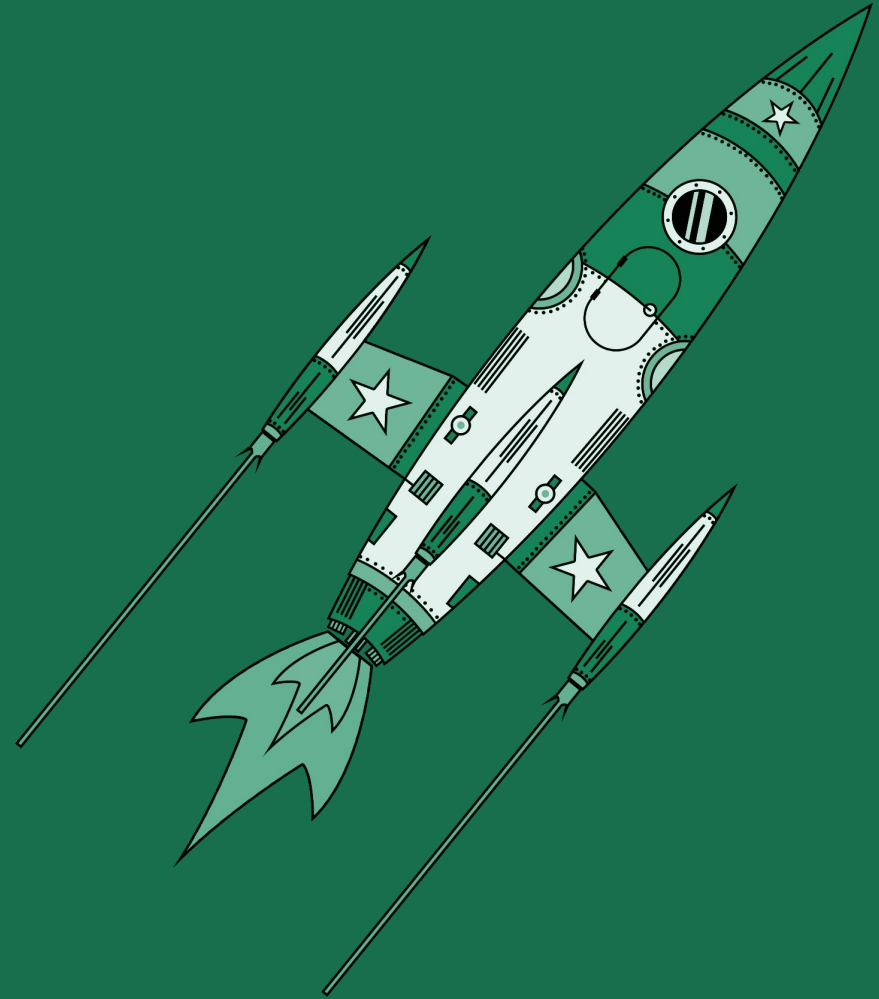
In your table groups, noting the guide times - remember to move on to (B):

- A) for 5 minutes: write down as many words or phrases as you can that relate to NCAS - think the research, tech, teams, sites, roles, aims, projects, etc.**
- B) for 15 minutes: choose a word/phrase per person from the words/phrases you listed in (A). Create a definition for it using only the 1000 most common words - you can discuss and work together or work through each individually. Write completed definitions on a sticky/post-it notes and when you have a sufficient batch, bring them to me at the front (I'll try to create a summary of all contributions). If you finish a word/phrase, move onto another one chosen from your list, or discuss with others, and repeat. Aim for spending ~5 minutes per word per person.**



Part 3 of 3

Our definitions



**National Centre for
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Review of our definitions

- Summarising contributions from the session...



Summary

- Up Goer Five is an XKCD web comic which describes parts of the NASA Saturn V rocket rocket in terms of only the 1000 most commonly used English words. The 'Up Goer Five Challenge' invites us to explain complex concepts or scientific research in the same limited vocabulary
- It challenges us to get to the very core of a concept, building communication skills and providing definitions built from the most basic building blocks which in theory can be understood by anyone
- We worked together list words related to NCAS and in turn to describe aspects of NCAS using the 1000 most common words, using tools to assist us (notably a validator)
- I will get the definitions we made together hosted on a webpage so we can view them at a later date and perhaps expand to include further words
- I hope you came away from this session having:
 - used your brain a bit and maybe your creativity too
 - an appreciation of the difficulty of explaining concepts without using technical jargon

